

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination (Backlog)

November 2013

CL204 Engineering Geology

Date: 28.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Sections I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain any two of the following. [04]

- i. Drainage pattern ii. Seismograph iii. Weathering

(b) Explain any two of the following landforms. [03]

- i. Alluvial fan ii. Blind valley iii. Pedestal rock

Q - 2 (a) Explain the Geological works done by the action of wind. [06]

(b) Describe the Classification of Earthquake. [04]

(c) Write the application of knowledge of engineering geology in the field of construction. [04]

OR

Q - 2 (a) Explain the methods of measurement of Seismic waves. [04]

(b) Describe the scope of Geology in different disciplines. [04]

(c) Briefly describe the internal structure of earth and methods of understanding the internal structure of Earth. [06]

Q - 3 (a) Write a short note on silicate structure and describe common silicate structures. [05]

(b) Write in detail about the structures seen in Igneous rocks. [05]

(c) Give example of any three metamorphic rock using in the civil engineering field. [04]
Explain the properties and use.

OR

Q - 3 (a) Describe rock cycle and discuss the way of formation of different type of rock types and the processes involved in it? [05]

(b) Explain the use of rocks and natural materials as a construction material [04]

(c) Write a short note on Hardness of mineral [05]

SECTION – II

Q – 4(a) Write short note on Geological mapping. Describe the characteristics of geological mapping and its importance for the civil engineering projects. [07]

Q - 5 (a) What are faults? How are they classified? Illustrate with neat sketch. [05]

(b) Which terminology is commonly adopted in describing folds and their effects? [05]

(c) Explain briefly (i) Columnar Joints (ii) Nonconformity [04]

OR

Q - 5 (a) Describe unconformities. How they are recognized in the field? [07]

(b) Write short note on recognition of fault and fold in the field. Write their engineering importance. [07]

Q - 6 (a) Discuss the following geological problem that are usually met with at dam / tunnel site (draw neat sketch if necessary) [07]

(i) Dams on shale formation

(ii) Dams on strata dipping downstream

(iii) Tunnel axis parallel to the dip direction

(b) Discuss relation of convergent and divergent plate boundaries and the stress imposed on rock masses (Draw neat sketch if necessary). [07]

OR

Q - 6(a) What does Landslide mean? State various factors affecting the landslide. [07]

(b) Write note on following geological characters that need to be investigated for engineering structures (draw neat sketch if necessary) [07]

(i) The depth of the bed rock (ii) The nature of the bed rock (iii) The structural disposition of rock